

Question

Calcium carbide reacts with N_2 at high temperatures to yield compound **A**. Hydrolysis of **A** produces an important pesticide intermediate **B**. Under alkaline conditions, **B** polymerizes to form dimer **C**. Heating **C** with dimethylamine hydrochloride and recrystallization yields **D**. Reaction of **B** with sodium cyanide and chlorine under alkaline conditions gives the sodium salt **E**. Researchers reacted 0.15 g $\text{Cu}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, 1.07 g **E**, and 0.52 g N,N-dimethylethylenediamine in an aqueous solution to obtain a dinuclear copper complex **F**, with an experimentally determined copper mass fraction of 22.55%. X-ray diffraction data reveal that ligand **E** in **F** exhibits two distinct configurations—monodentate coordination and bridging coordination—in a 1:1 ratio, with both configurations utilizing the same coordinating atom.

In the molecular structure of **F**, how many bonds separate the two farthest N atoms along the shortest path?

A. 2

B. 4

C. 6

D. 7

E. 8

F. 9

G. 10

H. 11

I. 12

J. 13

K. 14

L. 15

M. 16

N. 17

O. 18

P. 19

Answer

Correct Answer: **M**

Detailed Explanation

Based on the properties and reactions, it is inferred that:

A : $\text{Ca}(\text{NCN})$,

B : H_2NCN ,

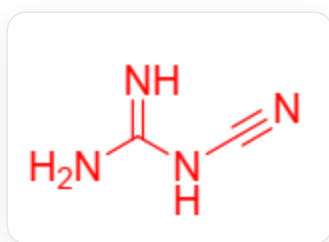
E : $\text{Na}[\text{N}(\text{CN})_2]$.

CHECKPOINT

1 PTS

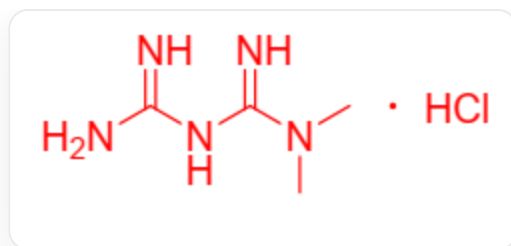
E : $\text{Na}[\text{N}(\text{CN})_2]$

and **C** :



C(=N)NC(=N)N

D :



The anion of **E** is abbreviated as dca, and N,N-dimethylethylenediamine is abbreviated as dmen.

The molar ratio is calculated as $\text{Cu} : \text{dca} : \text{dmen} = 1 : 2 : 1$

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1 : 2 : 1

Based on the mass fraction of copper, it is determined that **F** contains no water of crystallization,

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no water of crystallization

hence the chemical formula of **F** can be inferred as $\text{Cu}[\text{N}(\text{CN})_2]_2 (\text{C}_4\text{H}_{12}\text{N}_2)$

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2 PTS

$\text{Cu}[\text{N}(\text{CN})_2]_2 (\text{C}_4\text{H}_{12}\text{N}_2)$

F is a dimer, containing monodentate terminal and bridging $[\text{N}(\text{CN})_2]^-$ ligands. The two farthest N atoms originate from the two most distant N atoms on the monodentate ligands of the two Cu atoms.

The shortest path between the two farthest N atoms is:



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The farthest path is $\text{N}-\text{C}-\text{N}-\text{C}-\text{N}-\text{Cu}-\text{N}-\text{C}-\text{N}-\text{C}-\text{N}-\text{Cu}-\text{N}-\text{C}-\text{N}-\text{C}-\text{N}$

There are 17 atoms along this path, hence 16 bonds.

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1 PTS

The farthest distance is 16 bonds